1.36 Integration: Indefinite Integrals

Worked Solutions

1a)

$$4x^2 + x + c$$

1b)

$$2x^3 - 3x + c$$

1c)

$$\frac{5}{2}x^4 + 9x + c$$

1d)

$$\int (12x^3 + 20x^2 - 3x - 5)\, dx$$
$$= 3x^4 + \frac{20}{3}x^3 - \frac{3}{2}x^2 - 5x + c$$

2a)

$$\frac{1}{3}x^2 + \frac{1}{3}x^5 + c$$

2b)

$$\frac{1}{4}x^3 - \frac{2}{3}x^7 + c$$

2c)

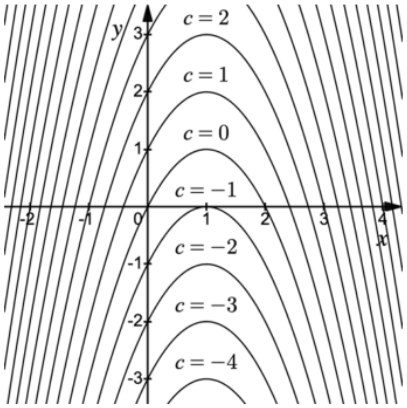
$$\frac{1}{20}x^8 - \frac{1}{16}x^6 + c$$

2d)

$$\int \left(\frac{2}{9}x^4 + \frac{14}{3}x^2 + 24\right)\, dx$$
$$= \frac{2}{45}x^5 + \frac{14}{9}x^3 + 24x + c$$

3)

$$y = 2x - x^2 + c.$$



4)

$$\frac{dy}{dx} = \frac{1}{6}x^2 + \frac{1}{6}x - 1$$

$$y = \frac{1}{18}x^3 + \frac{1}{12}x^2 - x + c$$

$$(6, 1) : 1 = 12 + 3 - 6 + c \Rightarrow c = -8$$

$$y = \frac{1}{18}x^3 + \frac{1}{12}x^2 - x - 8$$

5a)

$$y = \frac{3}{5}x^{5/3} + c$$

5b)

$$y = 5x^{6/5} + c$$

5c)

$$y = 12x^{7/4} + c$$

5d)

$$\frac{dy}{dx} = 4x^{1/3} + 12x^{1/6} + 9$$

$$y = 3x^{4/3} + \frac{72}{7}x^{7/6} + 9x + c$$

6a)

$$\frac{1}{4}x^4 + \frac{2}{5}x^{5/2} + c$$

6b)

$$-5x^{-1} - \frac{6}{5}x^{5/3} + c$$

6c)

$$-4x^{-2} - \frac{20}{11}x^{11/5} + c$$

6d)

$$\int (4x^{-8} - 12x^{-8/3} + 9x^{8/3})\,dx$$
$$= -\frac{4}{7}x^{-7} + \frac{36}{5}x^{-5/3} + \frac{27}{11}x^{11/3} + c$$

7)

$$\int (1 - x^{-4} - x^{-2} + x^{-6})\,dx = x + \frac{1}{3}x^{-3} + x^{-1} - \frac{1}{5}x^{-5} + c$$

8)

$$y = 30x^{4/3} - 2x^2 + c$$
$$(1, -1) : -1 = 30 - 2 + c \Rightarrow c = -29$$
$$y = 30x^{4/3} - 2x^2 - 29$$

9a)

$$y = 5x + \frac{2}{3}x^{-3} + c$$

9b)

$$\frac{dy}{dx} = 6x^{-2} - 12x^{-3}$$
$$y = -6x^{-1} + 6x^{-2} + c$$

9c)

$$\frac{dy}{dx} = 8x^{3/2} - 3x^{1/2}$$
$$y = \frac{16}{5}x^{5/2} - 2x^{3/2} + c$$

9d)

$$\frac{dy}{dx} = 4x^{-2} + 14x^{3/2} - 6x^{-7/2} - 21$$
$$y = -4x^{-1} + \frac{28}{5}x^{5/2} + \frac{12}{5}x^{-5/2} - 21x + c$$

10a)

$$\int (4x^{3/2} - 4x^{1/2})\,dx = \frac{8}{5}x^{5/2} - \frac{8}{3}x^{3/2} + c$$

10b)

$$\int x^{1/4}\,dx = \frac{4}{5}x^{5/4} + c$$

10c)

$$\int 2x^{1/5}\,dx = \frac{5}{3}x^{6/5} + c$$

10d)

$$\int 3x^{1/16}\,dx = \frac{48}{17}x^{17/16} + c$$

11)

$$\frac{dy}{dx} = 7x^{5/2} - 3x^2$$
$$y = 2x^{7/2} - x^3 + c$$
$$(1, -4) : -4 = 2 - 1 + c \Rightarrow c = -5$$
$$y = 2x^{7/2} - x^3 - 5$$

12a)

$$\frac{dy}{dx} = 4x^{-2} \text{ so } y = -4x^{-1} + c$$

12b)

$$\frac{dy}{dx} = 6x^{-3} \text{ so } y = -3x^{-2} + c$$

12c)

$$\frac{dy}{dx} = \frac{9}{5}x^{-4} \text{ so } y = -\frac{3}{5}x^{-3} + c$$

12d)

$$\frac{dy}{dx} = \frac{100}{9}x^{-10} \text{ so } y = -\frac{100}{81}x^{-9} + c$$

13a)

$$\int \frac{4}{3}x^{-1/2}\,dx = \frac{8}{3}x^{1/2} + c$$

13b)

$$\int 6x^{-1/3}\,dx = 9x^{2/3} + c$$

13c)

$$\int (8x^{-1/5} - 50x^{-1/3})\,dx = 10x^{4/5} - 75x^{2/3} + c$$

13d)

$$\int (x^{-2/3} - 6x^{-11/6} + 9x^{-3})\,dx$$
$$= 3x^{1/3} - \frac{36}{5}x^{-5/6} - \frac{9}{2}x^{-2} + c$$

14)

$$\frac{dy}{dx} = 8x^{-1/2}$$
$$y = 16x^{1/2} + c$$
$$(16, -8) : -8 = 64 + c \Rightarrow c = -72$$
$$y = 16x^{1/2} - 72 \Rightarrow y = 8(2\sqrt{x} - 9)$$

15)

$$\frac{dy}{dx} = 750x^{-4}$$
$$y = -250x^{-3} + c$$
$$(5, 248) : 248 = -2 + c \Rightarrow c = 250$$
$$y = -250x^{-3} + 250 \Rightarrow y = 250\left(1 - \frac{1}{x^3}\right)$$

16)

$$\frac{dy}{dx} = 6x^{-3/2}$$
$$y = -12x^{-1/2} + c$$
$$(8,0) : 0 = -3\sqrt{2} + c \Rightarrow c = 3\sqrt{2}$$
$$y = -12x^{-1/2} + 3\sqrt{2} \Rightarrow y = 3\left(\sqrt{2} - \frac{4}{\sqrt{x}}\right)$$

17)

$$\frac{dy}{dx} = 12x^{-1/3}$$
$$y = 18x^{2/3} + c$$
$$(27,0) : 0 = 162 + c \Rightarrow c = -162$$
$$y = 18x^{2/3} - 162 \Rightarrow y = 18(x^{2/3} - 9)$$

18)

$$\frac{dy}{dx} = x^{-7/2}$$
$$y = -\frac{2}{5}x^{-5/2} + c$$
$$\left(1, \frac{3}{5}\right) : \frac{3}{5} = -\frac{2}{5} + c \Rightarrow c = 1$$
$$y = -\frac{2}{5}x^{-5/2} + 1 \Rightarrow y = 1 - \frac{2}{5x^{5/2}} \Rightarrow y = 1 - \frac{2}{5x^2\sqrt{x}}$$

19)

$$\int \left(4x^{-5/2} - \frac{1}{4}x^{-3/2}\right) dx = -\frac{8}{3}x^{-3/2} + \frac{1}{2}x^{-1/2} + c$$

20)

$$\int \left(\frac{6}{5}x^{-7/2} - \frac{1}{6}x^{-8/3}\right) dx = -\frac{12}{25}x^{-5/2} + \frac{1}{10}x^{-5/3} + c$$

21)

$$\int \left(\frac{3}{4}x^{-7/2} - x^{7/6}\right) dx = -\frac{3}{10}x^{-5/2} - \frac{6}{13}x^{13/6} + c$$

22)

$$f'(x) = 9x^2 - 24 + 16x^{-2} + c$$
$$f(x) = 3x^3 - 24x - 16x^{-1} + c$$
$$f(2) = -2 : -2 = 24 - 48 - 8 + c \Rightarrow c = 30$$
$$f(x) = 3x^3 - 24x - \frac{16}{x} + 30$$

23a)

$$y = \frac{1}{5}kx^5 + c$$

23b)

$$y = \frac{1}{5}k^4x^5 + c$$

23c)

$$\frac{dy}{dx} = \frac{1}{k}x^{-4}$$
$$y = -\frac{1}{3k}x^{-3} + c$$

23d)

$$\frac{dy}{dx} = \frac{9}{16k^5}x^{-5}$$
$$y = -\frac{9}{64k^5}x^{-4} + c$$

24a)

$$\int (5x^{-2} + x)\,dx = -5x^{-1} + \frac{1}{2}x^2 + c$$

24b)

$$\int (3x^{-4} - 2x^{-2})\,dx = -x^{-3} + 2x^{-1} + c$$

24c)

$$\int (8x^{-2} - 9x^{-4})\,dx = -8x^{-1} + 3x^{-3} + c$$

24d)

$$\int \left(\frac{15}{7}x^{-2} - \frac{1}{2}x^{-5}\right) dx = -\frac{15}{7}x^{-1} + \frac{1}{8}x^{-4} + c$$

25a)

$$\frac{dy}{dx} = x - \frac{7}{5}x^{-3}$$
$$y = \frac{1}{2}x^2 + \frac{7}{10}x^{-2} + c$$

25b)

$$\frac{dy}{dx} = \frac{3}{4}x^{-5} - 2x^{-2}$$
$$y = -\frac{3}{16}x^{-4} + 2x^{-1} + c$$

25c)

$$\frac{dy}{dx} = \frac{1}{2}x^3 - \frac{2}{3}x^{-4}$$
$$y = \frac{1}{8}x^4 + \frac{2}{9}x^{-3} + c$$

25d)

$$\frac{dy}{dx} = \frac{1}{3}x^{-4} - \frac{1}{8}x^{-3} - \frac{1}{4}x^{-5}$$
$$y = -\frac{1}{9}x^{-3} + \frac{1}{16}x^{-2} + \frac{1}{16}x^{-4} + c$$

26)

$$f'(x) = x^{-2} - 8x^{-3}$$
$$f(x) = -x^{-1} + 4x^{-2} + c$$
$$f(1) = 0 : 0 = -1 + 4 + c \Rightarrow c = -3$$
$$f(x) = -x^{-1} + 4x^{-2} - 3 \Rightarrow f(x) = -3 - \frac{1}{x} + \frac{4}{x^2}$$
$$\Rightarrow f(x) = \frac{-3x^2 - x + 4}{x^2}$$

27)

$$f'(x) = \frac{196 - 196x + 49x^2}{x^{2/3}}$$
$$\Rightarrow f'(x) = 196x^{-2/3} - 196x^{1/3} + 49x^{4/3}$$
$$f(x) = 588x^{1/3} - 147x^{4/3} + 21x^{7/3} + c$$
$$f(1) = 462 : 462 = 588 - 147 + 21 + c \Rightarrow c = 0$$
$$f(x) = 588x^{1/3} - 147x^{4/3} + 21x^{7/3}$$
$$f(x) = 21x^{1/3}(x^2 - 7x + 28)$$

28)

At the turning point, $0 = 3 + k - 21$ so $k = 18$.

$$\frac{dy}{dx} = 3x^2 + 18x - 21 = 0 \text{ gives } x = 1,\ x = -7.$$
$$y = x^3 + 9x^2 - 21x + c.$$

At $(1, -6)$, $-6 = 1 + 9 - 21 + c$ so $c = 5$.

$$y = x^3 + 9x^2 - 21x + 5.$$

At $x = -7$, $y = 250$ so $(-7, 250)$.

29)

$$f(x) = \frac{1}{3}kx^3 + 37x^2 + 11x + c.$$

At $(-1, -100)$: $-126 = -\frac{1}{3}k + c$.

At $(-5, 0)$: $-870 = -\frac{125}{3}k + c$.

So $k = 18$, $c = -120$.

$$f(x) = 6x^3 + 37x^2 + 11x - 120.$$

30)

$$f(x) = 144x^4 + \frac{1}{3}ax^3 - 104x^2 + 40x + c.$$

$f\left(\frac{6}{5}\right) = 0$ gives $-\frac{275}{9} = \frac{125}{648}a + c$.

$f\left(-\frac{1}{2}\right) = 0$ gives $37 = -\frac{1}{24}a + c$.

So $a = -288$, $c = 25$.

$$f(x) = 144x^4 - 96x^3 - 104x^2 + 40x + 25.$$

31a)

$$(5 + x)^4 = x^4 + 20x^3 + 150x^2 + 500x + 625$$
$$(5 - x)^4 = x^4 - 20x^3 + 150x^2 - 500x + 625$$
$$(5 + x)^4 - (5 - x)^4 = 40x^3 + 1000x$$
$$\int (40x^3 + 1000x)\, dx = 10x^4 + 500x^2 + c$$

31b)

$$(3+2x)^5 = 32x^5 + 240x^4 + 720x^3 + 1080x^2 + 810x + 243$$
$$(3-2x)^5 = -32x^5 + 240x^4 - 720x^3 + 1080x^2 - 810x + 243$$
$$(3+2x)^5 - (3-2x)^5 = 64x^5 + 1440x^3 + 1620x$$
$$\int (64x^5 + 1440x^3 + 1620x)\,dx = \frac{32}{3}x^6 + 360x^4 + 810x^2 + c$$

31c)

$$(1-3x)^6 = 729x^6 - 1458x^5 + 1215x^4 - 540x^3 + 135x^2 - 18x + 1$$
$$(1+3x)^6 = 729x^6 + 1458x^5 + 1215x^4 + 540x^3 + 135x^2 + 18x + 1$$
$$(1-3x)^6 - (1+3x)^6 = -2916x^5 - 1080x^3 - 36x$$
$$\int (-2916x^5 - 1080x^3 - 36x)\,dx = -486x^6 - 270x^4 - 18x^2 + c$$

31d)

$$(1+2x)^7 = 128x^7 + 448x^6 + 672x^5 + 560x^4 + 280x^3 + 84x^2 + 14x + 1$$
$$(1-2x)^7 = -128x^7 + 448x^6 - 672x^5 + 560x^4 - 280x^3 + 84x^2 - 14x + 1$$
$$(1+2x)^7 - (1-2x)^7 = 256x^7 + 1344x^5 + 560x^3 + 28x$$
$$\int (256x^7 + 1344x^5 + 560x^3 + 28x)\,dx = 32x^8 + 224x^6 + 140x^4 + 14x^2 + c$$

32)

$$(4+2x)^3 = 8x^3 + 48x^2 + 96x + 64$$
$$(4-2x)^3 = -8x^3 + 48x^2 - 96x + 64$$
$$(4+2x)^3 - (4-2x)^3 = 16x^3 + 192x$$
$$\frac{dy}{dx} = 16x^3 + 192x \Rightarrow y = 4x^4 + 96x^2 + c$$
$$(-2, 1024) : 1024 = 64 + 384 + c \Rightarrow c = 576$$
$$y = 4x^4 + 96x^2 + 576 \Rightarrow y = 4(x^4 + 24x^2 + 144) \Rightarrow y = 4(x^2 + 12)^2$$

33a)

$$\frac{dy}{dx} = 3x + c_1, \qquad y = \frac{3}{2}x^2 + c_1 x + c_2$$

33b)

$$\frac{dy}{dx} = 2x^2 - 5x + c_1, \qquad y = \frac{2}{3}x^3 - \frac{5}{2}x^2 + c_1 x + c_2$$

33c)

$$\frac{dy}{dx} = 2x^3 - 4x^2 + 9x + c_1, \qquad y = \frac{1}{2}x^4 - \frac{4}{3}x^3 + \frac{9}{2}x^2 + c_1x + c_2$$

33d)

$$\frac{dy}{dx} = 2x^4 - \frac{7}{6}x^6 + c_1, \qquad y = \frac{2}{5}x^5 - \frac{1}{6}x^7 + c_1x + c_2$$

34a)

$$\frac{d^2y}{dx^2} = x^{1/2} \Rightarrow \frac{dy}{dx} = \frac{2}{3}x^{3/2} + c_1 \Rightarrow y = \frac{4}{15}x^{5/2} + c_1x + c_2$$

34b)

$$\frac{d^2y}{dx^2} = 5x^{-1/2} \Rightarrow \frac{dy}{dx} = 10x^{1/2} + c_1 \Rightarrow y = \frac{20}{3}x^{3/2} + c_1x + c_2$$

34c)

$$\frac{d^2y}{dx^2} = 3x^{-3} - \frac{1}{3}x^3 \Rightarrow \frac{dy}{dx} = -\frac{3}{2}x^{-2} - \frac{1}{12}x^4 + c_1$$
$$\Rightarrow y = \frac{3}{2}x^{-1} - \frac{1}{60}x^5 + c_1x + c_2$$

34d)

$$\frac{d^2y}{dx^2} = 8x^{-9/2} \Rightarrow \frac{dy}{dx} = -\frac{16}{7}x^{-7/2} + c_1 \Rightarrow y = \frac{32}{35}x^{-5/2} + c_1x + c_2$$

35)

$$\frac{d^3y}{dx^3} = 10x^{-3/2} \Rightarrow \frac{d^2y}{dx^2} = -20x^{-1/2} + c_1$$
$$\Rightarrow \frac{dy}{dx} = -40x^{1/2} + c_1x + c_2 \Rightarrow y = -\frac{80}{3}x^{3/2} + \frac{1}{2}c_1x^2 + c_2x + c_3$$

36)

$$f'(x) = 14x - 3x^2 + c_1$$
$$f'(3) = 2 : 2 = 42 - 27 + c_1 \Rightarrow c_1 = -13$$
$$f'(x) = 14x - 3x^2 - 13$$
$$f(x) = 7x^2 - x^3 - 13x + c_2$$
$$f(-3) = 2 : 2 = 63 + 27 + 39 + c_2 \Rightarrow c_2 = -127$$
$$f(x) = 7x^2 - x^3 - 13x - 127$$

37)

$$f'(x) = 12x^3 + 12x^2 + 12x + c_1$$
$$f'(1) = 5 : 5 = 12 + 12 + 12 + c_1 \Rightarrow c_1 = -31$$
$$f'(x) = 12x^3 + 12x^2 + 12x - 31$$
$$f(x) = 3x^4 + 4x^3 + 6x^2 - 31x + c_2$$
$$f(1) = 3 : 3 = 3 + 4 + 6 - 31 + c_2 \Rightarrow c_2 = 21$$
$$f(x) = 3x^4 + 4x^3 + 6x^2 - 31x + 21$$

38)

$$f''(x) = 18x^{-3}, \quad f'(x) = -9x^{-2} + c_1$$
$$f'(1) = -2 : -2 = -9 + c_1 \Rightarrow c_1 = 7$$
$$f'(x) = -9x^{-2} + 7$$
$$f(x) = 9x^{-1} + 7x + c_2$$
$$f(2) = 5 : 5 = \frac{9}{2} + 14 + c_2 \Rightarrow c_2 = -\frac{27}{2}$$
$$f(x) = 9x^{-1} + 7x - \frac{27}{2}$$

39)

$$f''(x) = 210x - 210x^{1/2} + 210x^{3/2}$$
$$f'(x) = 105x^2 - 140x^{3/2} + 84x^{5/2} + c_1$$

At $f'(0) = 0$: $0 = 0 - 0 + 0 + c_1$ so $c_1 = 0$.

$$f'(x) = 105x^2 - 140x^{3/2} + 84x^{5/2}$$
$$f(x) = 35x^3 - 56x^{5/2} + 24x^{7/2} + c_2$$

At $f(1) = 11$: $11 = 35 - 56 + 24 + c_2$ so $c_2 = 8$.

$$f(x) = 35x^3 - 56x^{5/2} + 24x^{7/2} + 8$$

40)

$$f''(x) = 12kx - 12, \qquad f'(x) = 6kx^2 - 12x + c_1$$

At $f'(1) = -15$: $-15 = 6k - 12 + c_1 \Rightarrow -3 = 6k + c_1$.

At $f'\left(\frac{1}{2}\right) = 0$: $0 = \frac{3}{2}k - 6 + c_1 \Rightarrow 6 = \frac{3}{2}k + c_1$.

So $k = -2$ and $c_1 = 9$.

$$f'(x) = -12x^2 - 12x + 9$$

So $f(x) = -4x^3 - 6x^2 + 9x + c_2$.

At $f(2) = -37$: $-37 = -32 - 24 + 18 + c_2$ so $c_2 = 1$.

$$f(x) = -4x^3 - 6x^2 + 9x + 1$$